Predicting Shipment ETA and Delay Propagation

Aiqon – Take-Home Assignment | Suraj

1. Problem framing and definitions

A shipment passes through six stages: **origin departure** → **ocean transit** → **berth** → **discharge** → **customs release** → **inland delivery**. Every stage has a scheduled and an actual timestamp. **Delay** at a milestone is *actual* – *scheduled* (hours); the target is the delay of final delivery, and a shipment is *late* if it exceeds 24h. Because ETA is not static, the system re-estimates it at four **checkpoints** (booking, departure, port arrival, customs release), using only information available at that moment. ETA is defined as

ETA = scheduled delivery + delay already incurred + predicted remaining delay

Predicting the *remaining* delay (rather than the total) is the key design choice: it forces the model to learn whether delay already on the books will be **absorbed** by downstream buffers or **amplified** by missed windows – i.e. to model propagation explicitly, and it makes explanations cleanly separable into "what has happened" (exact) and "what will happen" (model).

2. Data: a causal shipment simulator

Public datasets rarely link per-stage timestamps to their causes, so I built a mechanism-based simulator (`src/simulate.py`) over 6 real lanes (e.g. Shanghai→Los Angeles, Shenzhen→Rotterdam via Suez), 4 carriers with different schedule reliability, and 4 cargo types. Delays are never added to the final date directly; they arise from the mechanisms below and propagate through the timeline. 300 shipments are used for modelling plus 60 unseen "live" shipments for a hold-out test and the demo. Calibration: 51% of shipments are late (>24h), median delay 26h, heavy right tail – consistent with published container schedule-reliability levels of roughly 50–65%.

Stage	Mechanism (ground truth)	Propagation effect
Origin	Gamma delay ~ carrier reliability, origin congestion; rollover to next sailing (+7d), more likely in peak season	Source
Ocean	Schedule padding (~4%) eaten by weather; Suez/Panama disruption (+2–12d); carriers speed up to recover	Absorbs / shocks
Berth	Wait = f(congestion³); weekly AR(1) congestion per port; >24h late → berth window lost, back of queue	Amplifies (lateness × congestion)
Discharge	Yard productivity collapses above 60% congestion; volume effect	Amplifies
Customs	Docs incomplete +24h; inspection p = port × cargo × docs (+2–5d); no weekend processing	Calendar amplification
Inland	Fixed truck appointment: early release waits; late release must rebook (scarcer in peak)	Absorbs / amplifies

Ground truth is the actual delivery time; the simulator also stores the exact per-stage delay increments (they sum to the final delay), which lets me *validate explanations* – impossible on real data. To mimic EDI/AIS gaps, 5–15% of intermediate events (arrival, berth, discharge, customs release) are removed in the observed dataset.

3. Features

- **Static (booking):** carrier reliability, origin congestion, forecast destination congestion and weather (noisy forecasts, not truth), chokepoint exposure, voyage/inland length, volume, peak season, *inspection risk* = port strictness × cargo risk × documentation, scheduled release on a weekend.
- **Propagation / interaction features:** departure and arrival delay, delay accrued at sea, *missed berth window* (>24h late), *lateness* × *congestion*, early-arrival (absorbed at anchor), *projected clearance hitting a weekend* (hours of working time before Saturday), *slack to the truck appointment* and *missed appointment*.
- **Missing data:** a missing arrival event is reconstructed from the berth time minus the expected congestion-dependent wait (flagged); remaining gaps are left as NaN, handled natively by the trees.
- Routes are described by their attributes (length, chokepoint, inland leg) rather than one-hot IDs, so the model can generalise to a new lane.

4. Models and evaluation protocol

Per checkpoint: a gradient-boosted tree regressor (scikit-learn HistGradientBoosting, absolute-error loss, depth 3, chosen for small data, non-linearity and native NaN support) predicts remaining delay; a classifier gives P(late >24h); quantile models with **conformal calibration** (CQR) give an 80% ETA band. Baselines: schedule (assume on time), carry-forward of the current delay, carry-forward + route median, and a ridge regression on the same features. Evaluation: 5-fold CV repeated 3× (MAE in hours, ROC-AUC, interval coverage) plus the untouched 60-shipment hold-out.

5. Results

Checkpoint	Schedule	Carry-fwd	+Route med.	Ridge	GBM (ours)	Hold-out GBM	AUC	80% cover.
Booking	63.0	63.0	61.8	64.3	**56.9**	51.1	0.74	82%
Departed	63.0	48.4	48.9	52.0	**44.2**	44.2	0.78	78%
Port arrival	63.0	44.6	38.2	31.8	**29.2**	29.8	0.85	78%
Customs released	63.0	10.7	9.7	8.8	**5.6**	5.9	0.99	83%

Table 1 – MAE of final delivery delay (hours), CV mean; hold-out on 60 unseen shipments.

The model beats every baseline at every checkpoint. At booking little is knowable (MAE 57h vs 63h for the schedule) – random shocks such as rollovers and canal disruptions dominate, and the honest output is a wide band. Once the vessel reaches port, the model is **34% better than carrying the current delay forward** (29h vs 45h), because it knows when that delay will be amplified at the berth or in customs. After customs release it is 47% better (5.6h). The ridge model, given the same interaction features, is a respectable second – evidence that most of the value lies in the feature design. Conformal bands achieve close to the nominal 80% coverage in CV.

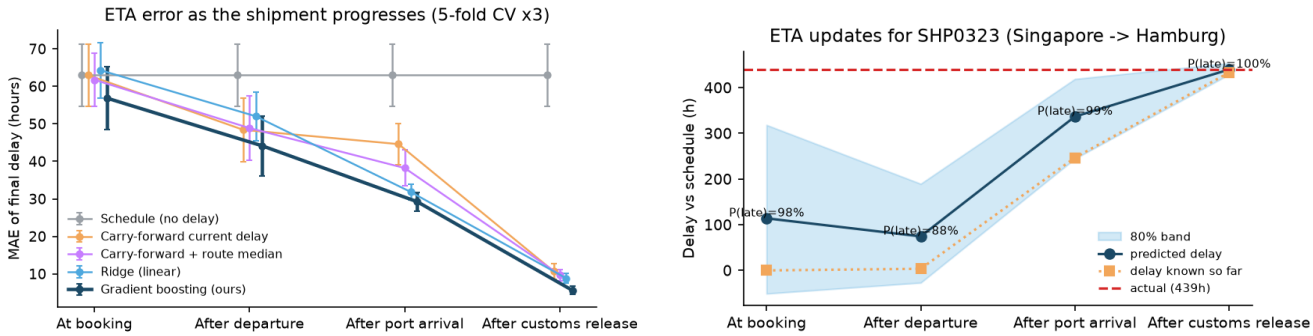


Figure 1 – Left: error vs checkpoint. Right: ETA updates for an unseen Singapore→Hamburg shipment that suffered a Suez diversion; the band narrows and converges on the actual delivery.

6. How delays propagate

Table 2 measures the delay added *after* the vessel reaches port (final – arrival delay). At low congestion a vessel arriving up to 24h late is largely **absorbed** (median -1h; 52% of shipments recover). At high congestion the same small lateness adds **+48h**, and a >24h-late vessel (missed berth window) adds **+108h** vs +34h at low congestion – a ~3× amplification. Figure 2 (right) shows the model learned exactly this non-linear interaction: a step at the 24h berth-window threshold whose height grows with congestion. Figure 3 (right) shows where delay originates per lane: customs and origin departure dominate on the trans-Pacific lanes, berth congestion is largest into Hamburg, and Suez diversions add a large ocean-transit share on Nhava Sheva→Felixstowe.

Congestion	Arrival	n	Med. arrival delay	Med. final delay	Med. added downstream	Absorbed
low (<0.5)	early/on time	45	-8h	2h	+15h	0%
low (<0.5)	late 0-24h	21	7h	6h	-1h	52%
low (<0.5)	late >24h	10	163h	231h	+34h	0%
medium (0.5-0.7)	early/on time	87	-9h	1h	+16h	1%
medium (0.5-0.7)	late 0-24h	44	8h	22h	+9h	41%
medium (0.5-0.7)	late >24h	25	135h	198h	+65h	0%
high (>0.7)	early/on time	29	-8h	37h	+44h	0%
high (>0.7)	late 0-24h	28	10h	52h	+48h	25%
high (>0.7)	late >24h	9	149h	312h	+108h	0%

Table 2 – Downstream delay added after port arrival, by arrival lateness × congestion.

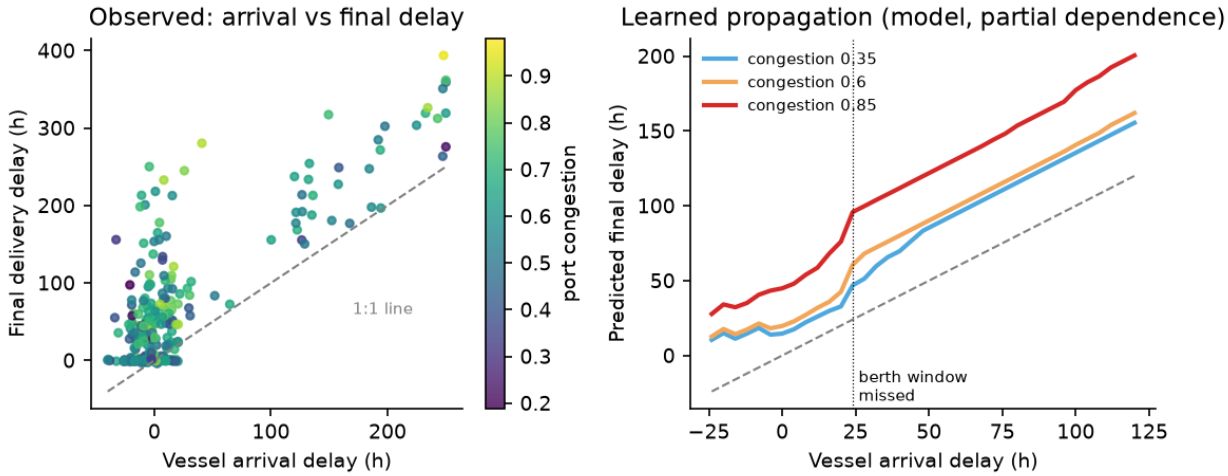


Figure 2 – Left: observed arrival vs final delay coloured by congestion. Right: model partial dependence; the gap above the dashed 1:1 line is delay added downstream.

7. Explanations

Each prediction is explained in two layers: (i) **delay already incurred**, decomposed exactly into milestone-to-milestone increments (missing events merge adjacent stages), and (ii) **expected further delay**, explained with TreeSHAP values aggregated into six plain-language cause groups. Example (unseen shipment, at port arrival, from `python predict.py`):

```
Shipment SHP0323  |  Singapore -> Hamburg  |  carrier Meridian  |  cargo perishable  |  checkpoint: C2_port_arrival
  Scheduled delivery : 2025-10-21 11:12
  Predicted ETA      : 2025-11-04 11:54   (delay +14.0d, 80% band +10.2d .. +17.5d)
  P(delay > 24h)     : 99%

  Delay already incurred: +10.3d
     origin (until departure)                  +4h
     departure -> port arrival               +10.1d

  Expected further delay: +3.8d  (typical shipment at this stage: +33h)
  Main drivers vs typical shipment:
     Destination port congestion         +36h   [lateness x congestion score=193, port congestion on arrival=0.78]
     Upstream delay (propagation)        +12h   [vessel arrival delay=10.3d, early arrival (absorbed at anchor)=0h]
     Customs & documentation             +11h   [documentation completeness=no, working hours before weekend=62]

  ACTUAL: delivered 2025-11-08 18:24, delay 439h
```

Are the explanations right? Because the simulator records the true cause of every hour of delay, I checked, for the 154 shipments with >24h still to come at port arrival, whether the top SHAP group (port congestion / customs / inland) matches the stage that truly generated the most excess delay. Agreement is **71%** (top-2: 88%) vs 33% by chance. Figure 3 (left) shows how the dominant drivers shift over the lifecycle: customs and port congestion matter most early on; after release only truck-appointment slack and the propagated delay remain.

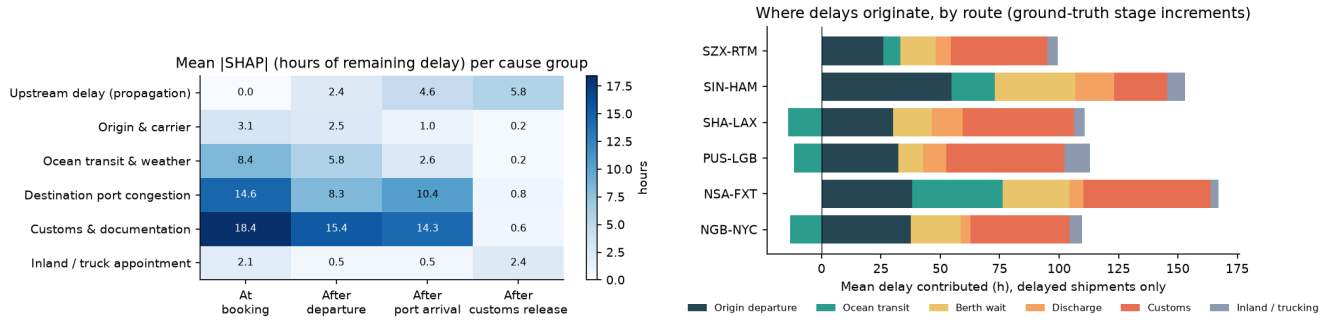


Figure 3 – Left: mean |SHAP| (hours) per cause group and checkpoint. Right: ground-truth origin of delay per lane (delayed shipments).

8. Robustness, assumptions and limitations

- **Missing events:** tripling the missing-event rate (up to 45% of berth/discharge events) moves MAE at port arrival only from 28h to 29h and after customs from 4.5h to 10.8h, staying well below the carry-forward baseline.
- **Assumptions:** schedules are fixed at booking; congestion is a weekly port index available from a port/AIS feed; customs works weekdays only; one truck appointment per shipment; forecasts at booking are noisy versions of the truth. All coefficients are stylised and documented in `src/config.py`.
- **Limitations:** the data is synthetic, so absolute errors are not real-world claims; the value shown is the method and its validation. With 300 records the uncertainty of each metric is visible in the CV spread (Fig. 1); the 60-shipment hold-out coverage after customs (73%) shows that bands are less reliable when very narrow.
- **Next steps with real data:** AIS vessel positions for live transit delay, port call/berth line-ups for congestion, customs-broker events; a sequence model across many vessels to capture network effects (one late vessel delaying the next ones at the same berth); and causal “what-if” estimates (e.g. value of expediting documents) on top of the explanations.